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Letter of Collaboration

This is a letter of collaboration for the Academy Professor project proposal “Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)” of Assoc. Prof. Maarit Korpi-Lagg from Aalto University. I am a Professor of Parallel and Distributed Data Science, specializing in parallel computing methods for big data processing and parallel GPU algorithms. I am very interested in collaboration on GPU-accelerated stencil algorithms, as efficient concurrency control of GPU computing is one of my key research areas. We have already done a joint publication “SOMA: Observability, monitoring, and in situ analytics for Exascale applications” which focused on the analysis on simulation data from massive GPU based HPC simulation using in-memory data streaming between the main simulation and a data analysis application opportunistically using computing CPU resources unused by the GPU simulation. I am very interested in continuing our collaboration also in the context of this new MomEnt proposal, where our main focus will be integration and optimization of the big data analysis workflows to LUMI supercomputer.

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